

Robotics 19/02/2021

Points: -/270

Kinematics

The robot in the picture is a differential robot for logistics and its name is Jobot. It has a lidar sensor under the loading plate and three sonars in the front.



1

What are direct and inverse kinematics (define them), and what are they used for.
(20 Points)

The kinematics is the study of the robot's motion without considering its forces, thus we just consider its geometric relationships.

The direct kinematics have prior knowledge of the control variables ($v(t)$, $\omega(t)$) and their goal is to find the robot pose ($x(t), y(t), \theta(t)$).

The inverse one instead have the robot's pose and look for the control variables.

2

Discuss the direct kinematics for the Jobot (formulas and description)
(20 Points)

Since the Jobot has differential drive kinematics we have the wheel's speed V_l and V_r (respectively the left wheel's velocity and the right speed velocity) and the angular velocity ω . The linear velocities are different for each wheel since we can control them separately. Under the assumption of no slippage and full rotation without translation of the wheels we have a single ICR (Instantaneous Center of Rotation) when the difference of speed between the two wheels is different from zero, otherwise the ICR does not belong to the plane with the radius that tends to infinity.

The formulas for the control parameters are as follows:

$$V_r = \omega(R + L/2)$$

$$V_l = \omega(R - L/2)$$

$$\omega = (V_r - V_l)/L$$

$$V = (V_r + V_l)/2$$

$$R = L/2 \cdot (V_r + V_l) / (V_r - V_l)$$

$$\text{ICR} = (x - R \cdot \sin(\theta), y + R \cdot \cos(\theta))$$

with V = robot's velocity, ω = angular velocity and R = radius of the ICR

The formulas to evaluate the direct kinematics are:

$$x(t) = \int v(t) \cos(\theta(t)) dt$$

$$y(t) = \int v(t) \sin(\theta(t)) dt$$

3

Discuss the inverse kinematics for the Jobot
(20 Points)

By having the control parameters V and w we can apply a simple algorithm to reach the desired goal:

- 1) turn the wheels such that they point towards the desired goal
 $-V_l(t) = V_r(t) = V_{max}$
- 2) move straight to the goal
 $V_l(t) = V_r(t) = V_{max}$
- 3) rotate again to reach the desired orientation (and final pose)
 $-V_l(t) = V_r(t) = V_{max}$

Sensors and actuators

We want to perform odometry for the robot in the previous exercise.

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What is odometry?
 (10 Points)

Odometry is the estimation of the robot pose by the integration of its control variables. An example has been already demonstrated in question 1. If we don't take into account the possible error in the estimation (due to slippage, baseline error, etc) we are performing the least precise kind of odometry, called dead reckoning. A way to correct this problem are the

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How does an incremental rotary encoder work and what is it used for?
 (10 Points)

Rotary encoders are a way to estimate the robot's wheels current position. They use phototransistors that sense through a holed shaft the light emitted from diodes to convert the wheel's motion into electronic pulses.

The main difference between incremental and rotary encoders is that the former use the rotary motion of the wheels to understand the position, while the latter can even use the still position to understand the current angular position of the wheels. Absolute encoders are more expensive and more precise (their resolution can reach $360^\circ/2^N$, where N is the number of

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How does an absolute rotary encoder work and what is it used for?
(10 Points)

the answer above works for this one as well

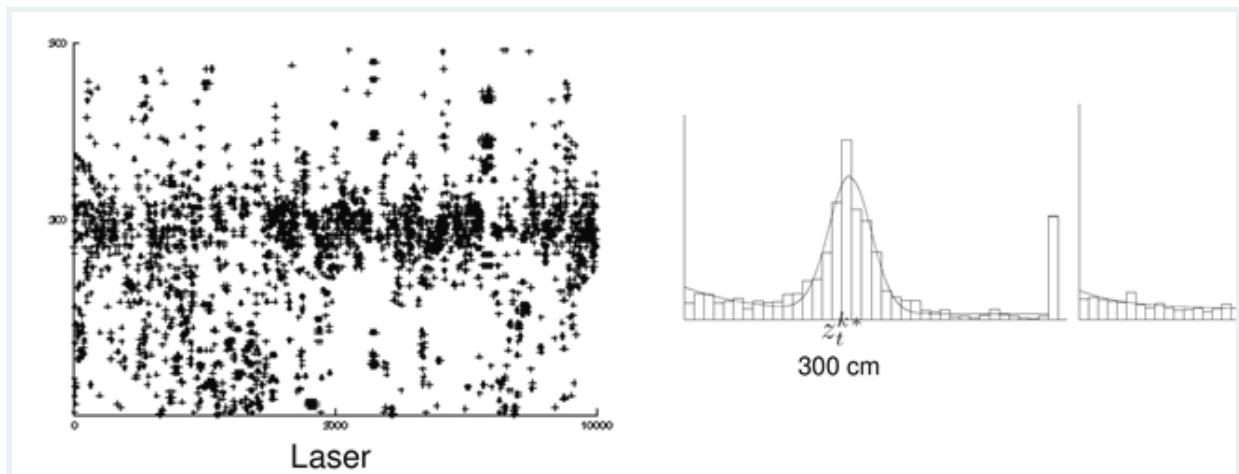
7

What is the quadrature technique?
(10 Points)

It's a technique used in incremental rotary encoders to increase the encoding resolution by shifting by a 1/4 of period the second square wave of electronic pulse coming from the diode. In this way every time there's a change of state between one of the two waves we can evaluate it as a smaller angle shift of the wheels.
It's a clever way to improve the encoder quality without changing drastically its structure and

Simultaneous Localization and Mapping

Consider again the Jobot, it is equipped with lidar. The following pictures refer to the beam sensor model of a lidar.



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What do the three plots in the picture represent?
(30 Points)

The first plot represents the data acquired from the lidar. It is a collection of high energy beams reflected in a single spot and recollected afterwards. This technique ensures a high pass filter on a lot of the background noises (EG: other lights, unwanted reflections,...) that can happen during a scan.

The two following graphs are two beam sensor models taken from the scan, one where the target is at a distance of 300 cm, the other when the target is at 400 cm.

The models are a combination of 4 different models:

- a gaussian model, that represents the target ideally measured
- a reverse exponential, that represents unexpected obstacles in the way of the target (the exponential is cut right after the target position)
- a uniform distribution, that represents random measurements
- a Dirac distribution, that represents the max range the laser can reach if no targets are measured

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How can you use a beam sensor model for localization using a discrete Bayesian filter?
(20 Points)

The Bayesian filter is the step that holds the sensor model and the motion model together. It calculates a belief ($Bel(x)$) of the position, which is a probabilistic estimate of the robot position. First step I would discretize the control space of the robot as multidimensional grid. For each robot state I will have a piecewise constant approximation. Then I would turn the continuous distribution of the $Bel(x)$ into a histogram.

At the first observation through a measurement I'd weigh the data accordingly and the histogram will have some peaks in the most probable position.

As soon as the robot moves I'd propagate the histogram.

By alternating the two steps of movement and observation the histogram will tend to peak in the most likely robot's position.

The beam sensor model is a fine fit for the model discrete filter since it peaks along the median

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Which algorithm for SLAM would you suggest for this robot? Why?
(20 Points)

Since the robot can turn on the spot a VFH can be implemented, but the Jobot works in a warehouse, thus multiple other robots and obstacle may occur during its travel, thus it could be better a VFH+. My choice is based on the increased obstacles dimension of the algorithm that could help in reaching the spot without accidents, and the curves that could be performed by the robot due to shift. The algorithm would be ideal based on the hypothesis of large enough corridors and not too many obstacles that could occlude the path and cause the reach of local

Robot Operating System

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Why does the movebase navigation package (robot navigation package) use actions instead of standard nodes or services?
(30 Points)

The movebase package is the main element of the ROS navigation package. It uses the principal planning and control functions coming from the plugin of the dynamic conf. Since it's the main functioning node of the packet uses actions because have longer execution time (thus they can perform more complex tasks) and can be monitored constantly. Plus the actions can be cancelled at any time, whereas services cannot. With actions we have the certainty of its correct execution, which is crucial to a node of such importance. Moreover

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What is a rosbag and what it is used for?
(20 Points)

The rosbag package is used to record, collect and reproduce scan recordings made by proprioceptive or eteroceptive sensor attached to the robot. A common usage after the recording is the replay of its collected informations (with the command rosbag play) in order to reproduce at each timestep the collected values and create a proper simulation of what happened at the robot and its sorroundings.

Motion Control

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With a differentail drive robot, do we really need DWA? Why?
(20 Points)

Not really, since DWA takes into account only circular trajectories and differential drive robots can perform both linear and circular ones. A VFH (or VFH+) is more than enough to use on such robots, since they can turn in place by using opposite speeds on each of their wheels. Another important factor to take into account is the speed required to calculate the trajectories

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With an Ackerman robot, do we really need DWA? Why?
(20 Points)

It's a different story for the Ackermans, since their curving capacity is limited and they only can turn by turning their front wheels and by performing a curve. Moreover is a nice combination the failsafe procedure to prevent bumps into obstacle implemented into DWA paired with the dynamic window to choose from the optimal speed for the robot. Since typically Ackerman vehicles are bulkier they can have a heavier payload, thus they can afford a "heavier" computing algorithm like DWA. It's a standard procedure to pair DWA and Ackermans because DWA is the standard obstacle

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What is the navigation function used by DWA with trajectory rollout?
(10 Points)

The nav function is the same used by DWA, but with less possible trajectories to consider since the algorithm discards each sampled velocity which would result into an illegal trajectory.

The global navigation function goes as follows:

$$NF = \alpha \cdot vel + \beta \cdot f + \gamma \cdot D_{nf} + \gamma \cdot goal$$

with:

- α, β, γ and δ as tuning parameters
- vel as the forward robot velocity
- nf as the cost to reach the goal
- D_{nf} as cost to follow the global path
- $goal$ as the nearness to the goal

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